Trustworthy G1 Manipulation for VLA Research

I built a physically audited Unitree G1 manipulation testbed in MuJoCo, then used it to study the interface between classical robot control and future vision-language-action learning.

Entrance-test project for graduate lab selection

2026-09-04

I Treated the Entrance Test as a Research System, Not a Demo

Host reality

Isaac Lab was unavailable on Intel macOS, so I used the official Unitree MuJoCo path and recorded the deviation instead of hiding it.

Physical validity

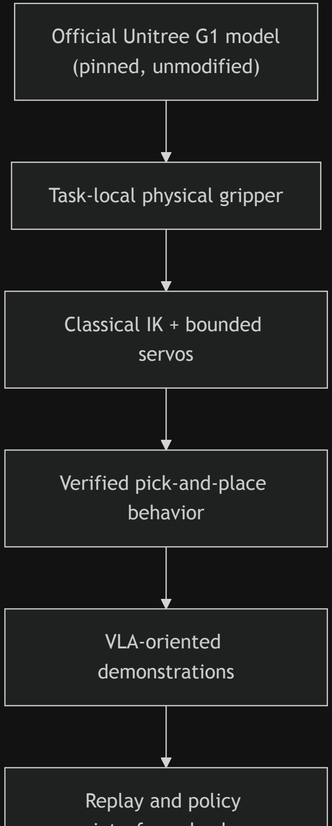
Every success had to come from contact and actuation: no cube weld, attachment, teleport, mocap assistance, or post-reset qpos manipulation.

Research traceability

I kept phase reports, logs, tests, videos, dataset schemas, and explicit failure records for each design decision.

The project became a small, evidence-driven robotics platform for asking VLA-ready manipulation questions.

The Core Claim: I Can Build the Layer Between Physics and Learning



This is the part many learning projects assume already exists: a trusted embodiment, real contact, repeatable evaluation, and a clean boundary between policy observations and privileged simulator state.

What Was Delivered Within the Time Box

Area	Delivered evidence
Simulator setup	Official `unitree_mujoco` G1, pinned commit, MuJoCo smoke tests
Embodiment audit	29 actuators mapped; wrist, sites, contacts, and G1 variants inspected
Manipulation interface	Task-local physical parallel gripper on the right wrist
Task 1	Red cube pick-and-place to blue target, fixed-base G1, real contacts
Task 2	Language-conditioned object selection with red/green cubes
Dataset	32 episodes, 28 successes, RGB + proprioception + actions + privileged split
Evaluation	311 tests, 0 unexpected failures; limitations reported rather than hidden

Not attempted: Task 3, model training, and model inference. The deck does not claim them.

I Chose a Fixed-Base G1 Because the Measurement Demanded It

Measured before implementation

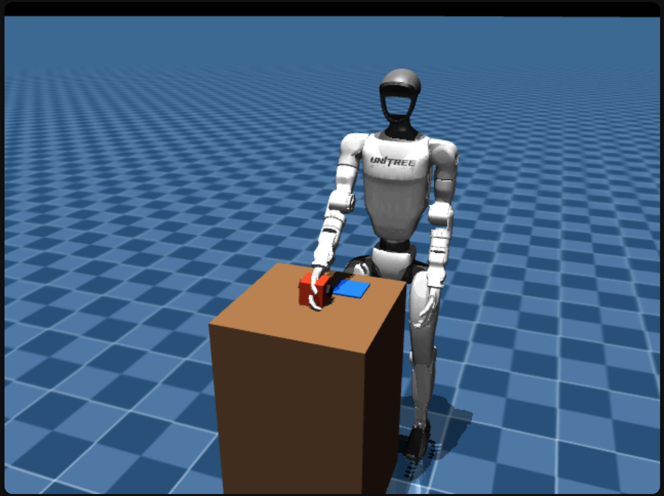
- The free-standing model drifted **0.93 m in 2 s** under small arm torques.
- The stock G1 variants contained no actuated gripper, Inspire hand, or Dex3 hand.
- The stock model had no wrist site suitable as a direct IK target.

Design choice

- Use the official **29-actuator G1** body and arm model.
- Add a **task-local physical gripper** under `right_wrist_yaw_link`.
- Fix the pelvis and torso for the first manipulation MVP.
- Keep all vendor files unmodified.

This made the task smaller, but also more honest and testable.

The Gripper Was Physical, Local, and Audited



- Two actuated slide-joint fingers.
- Collision-enabled pads with bilateral contact checks.
- A documented TCP site for IK and replay.
- Cube initialization allowed only before the first physics step.
- No object assistance after the trial starts.

The central technical promise is simple: if the cube moves, it moved because the simulated robot touched it.

A Failed Controller Became the Research Turning Point

Attempt 1

Torque PD + DLS IK failed after three documented tuning attempts.

Diagnosis

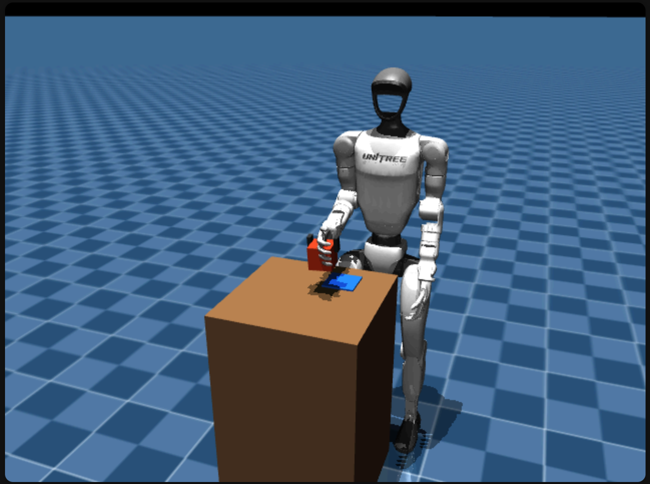
Shoulder/elbow and wrist torque authority differed by about **5x**, so one shared gain design was structurally wrong.

Replacement

Bounded MuJoCo position servos + waypoint IK produced deterministic grasp and lift.

The contribution was not just tuning until something worked. It was knowing when the representation itself had to change.

Task 1 Works Inside a Measured Envelope



Metric	Result
Nominal pick-and-place	Deterministic success, 5/5
Reachable setup variants	3/3 succeed
Original five-position coverage	3/5
Human visual acceptance	Prototype completed with limitation
Strict 10 mm grasp-slip bar	Not met: 25.92 mm

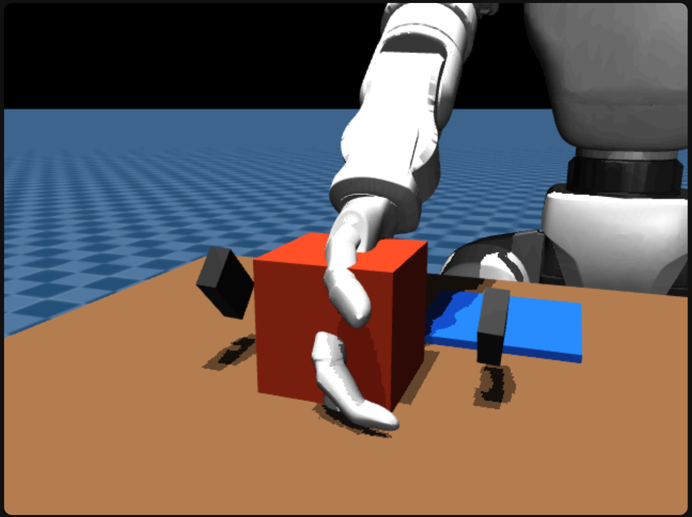
I report the boundary: task-level behavior is reliable in the reachable envelope, while strict grasp quality remains open.

The Best Evidence Was a Bug the Tests Missed

After the tests were already green, visual review showed a decorative vendor hand mesh clipping through the cube.

That did not invalidate the physical contact result, because the mesh was collision-free, but it did invalidate the presentation quality of the evidence.

I removed the misleading visual mesh in the task-local scene and kept the physical audit boundary intact.



This is the kind of failure that only appears when engineering tests and human evidence review are both taken seriously.

The Dataset Was Designed for Future Learning, Not Just Logging

Policy-facing observations

- Head/torso RGB camera at 10 Hz
- Robot joint positions and velocities
- TCP pose
- Gripper command/state

Privileged expert/evaluation state

- Cube and target poses
- Contact state
- State-machine phase
- Object selection metadata

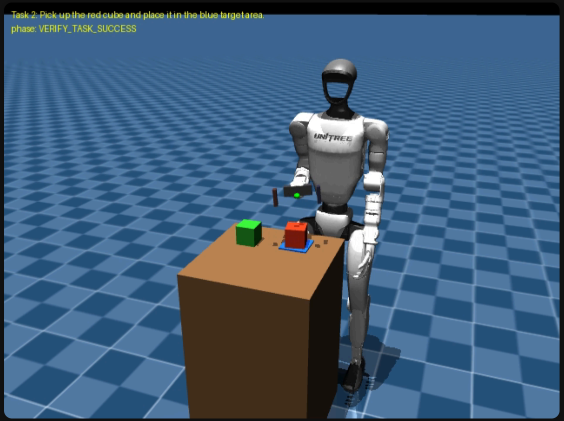
> The important design habit: privileged state is useful for evaluation, but it must not quietly become a learner input.

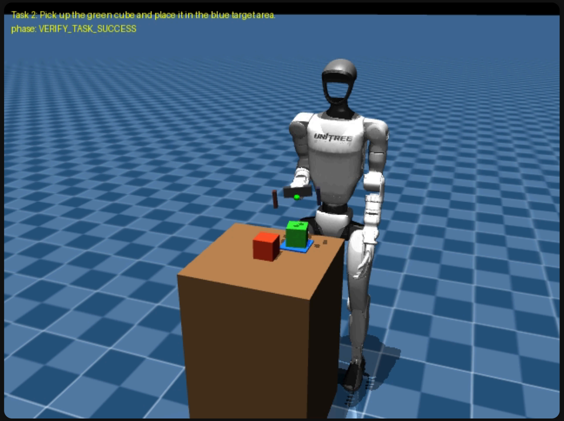
Replay Fidelity Exposed an Action-Representation Problem

Replay representation	What it tested	Result
10 Hz zero-order hold	Naive action interface	48.7 mm TCP error
Exact 500 Hz execution	Simulator determinism	~1e-5 m scale error
Static 10 Hz phase goal	Coarse goal representation	~97 mm TCP error
H=5 chunks at 50 Hz	Shipped policy-action representation	**8.09 mm / 5.99 mm prototype errors**

The simulator was not the problem. The action interface was.

Task 2 Shows the First Step Toward Language-Conditioned Manipulation





The scene contains two physically identical cubes. The instruction selects which object should be moved to the shared blue target.

- 4 configurations × 3 deterministic trials = **12/12 pass**
- **0** wrong-object placements
- Distractor displacement stays within **0.0–1.73 mm**
- This is task-conditioned control, not learned language grounding

The Most VLA-Relevant Result Is the Clean Boundary

What already exists

- Physical manipulation environment
- Language-style task specifications
- RGB/proprioceptive observations
- Demonstration trajectories
- Replay checks and dataset validation

What remains research work

- Replace privileged object ID with visual-language grounding
- Train a baseline behavior cloning policy
- Vary target locations and object classes
- Test generalization beyond deterministic repeats

This is a credible starting point for VLA research because the non-learning substrate is already measurable.

Why This Project Fits a Robotics AI Lab

Robotics

I can build a contact-rich manipulation task, diagnose embodiment limits, and keep physical validity constraints explicit.

ML systems

I think in observation/action schemas, replay fidelity, leakage boundaries, and reproducible datasets.

Research

I do not hide failed hypotheses. I use them to decide which abstraction should change next.

The Next Study Is Clear and Small Enough to Execute

Next phase	Acceptance criterion
Behavior cloning baseline	Train/evaluate on Task 1 demonstrations; report success and failure modes
Grounded object selection	Replace privileged `selected_object_id` with RGB + instruction input
Target variation	Move the blue target in scene geometry and prove replay/camera consistency
Grasp quality repair	Reduce true grasp-phase slip below 10 mm without scripted assistance
Task 3	Add a new manipulation primitive only after the above are stable

The research direction is not "make the demo bigger." It is "remove one privileged assumption at a time."

What I Want to Contribute

I want to join the lab to work on robot learning systems where VLA policies are evaluated through real physical interaction, clean data interfaces, and honest failure analysis.

This project shows my current level: early in robotics, but already able to build a testbed, debug it quantitatively, and turn it into research questions a lab can continue.

Appendix

Evidence for Technical Questions

Appendix A1 — Requirement Coverage

Requirement	Status	Evidence
Unitree G1 in simulation	Complete	Official G1 in MuJoCo
Isaac Lab	Deviation	Not supported on Intel macOS
Task 1	Complete	Pick-and-place, fixed-base, physical gripper
Task 2	Partial/complete within scope	Object-conditioned selection, not language grounding
Task 3	Not attempted	Time-boxed out
Model-free environment control	Complete	Classical IK + bounded servos
VLA-oriented data pipeline	Complete	RGB/proprioception/actions/privileged split
Existing policy inference	Not attempted	No model run

Appendix A2 — Physical Integrity Rules

- Vendor `unitree_mujoco` files remain unmodified.
- G1 model is pinned at commit `4134cb5dc7ff1ba7f484deda48b5274b58694519`.
- Cube free-joint state may be initialized only before the first physics step.
- After a trial starts: no direct cube qpos/qvel write, no weld, no attachment, no teleport, no mocap manipulation, no applied cube force.
- A guard and automated tests enforce this boundary.

Appendix A3 — Main Limitations

- Strict max 3D grasp-slip target is not met: **25.92 mm** versus 10 mm.
- Two of five original cube offsets are outside the reachable envelope.
- Target location is fixed in the current dataset.
- Task 2 uses privileged object selection metadata; it does not perform learned language grounding.
- Seven of 29 scaled policy-action replays exceed 10 mm, all first diverging after release during RETREAT.
- No model training or inference was attempted.

Appendix A4 — Source Evidence

- Main report: `submission/entrance_test_report.md`
- Machine-readable summary: `submission/results_summary.json`
- Reproduction guide: `submission/REPRODUCE.md`
- Dataset card: `submission/DATASET_CARD.md`
- Phase reports: `reports/phase*.md`
- Logs: `logs/*.json`
- Evidence media: `artifacts/` and `submission/videos/`
- Slide assets: copied from real evidence stills into the Slidev `public/` folder

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